



Meaningful oversight of medical AI beyond human in the loop

Davy van de Sande, Nicoleta Economou-Zavlanos & Michel E. van Genderen



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Human oversight of medical AI is increasingly required, but clinician presence alone does not make oversight meaningful. We propose four interlocking conditions—epistemic capacity, cognitive space, decisional authority, and intervention effectiveness—that determine whether human judgment can function as a safety mechanism. Embedding these conditions across procurement, deployment, monitoring, and decommissioning can turn oversight into an operational patient-safety capability across predictive, generative, semi-autonomous, and agentic systems.

Artificial intelligence (AI) increasingly informs diagnosis, triage, risk stratification, and treatment, yet many health systems still lack practical ways to govern it. Recent regulatory efforts, including the European Union's AI Act, have placed renewed emphasis on meaningful human oversight in high-risk medical AI¹. However, clinician presence in the workflow does not, by itself, make oversight meaningful². Health systems need a way to determine whether oversight is informed, feasible, institutionally supported, and capable of changing the course of AI-mediated care.

This argument does not rest on the assumption that human judgment is categorically superior to AI. In bounded or well-specified tasks, AI may equal or exceed clinician performance³. Governance challenges emerge when such systems are embedded in clinical workflows where uncertainty, actionability, accountability, reversibility, and downstream consequences matter alongside accuracy. The relevant oversight question is therefore not whether AI can be useful, but how its use should be structured when translated from bounded evaluation settings into clinical workflows. Meaningful oversight is therefore not a rejection of AI capability, but a condition for translating that capability safely into care.

This challenge intensifies as AI moves from recommendation to action. Our proposed framework applies across a continuum of medical AI systems: predictive tools that estimate risk or suggest actions, generative tools that draft or summarize clinical content, semi-autonomous workflow systems, and agentic systems that may initiate outreach, schedule visits, retrieve information, route messages, or trigger tasks with limited real-time human input^{4,5}. In conventional decision support, oversight often centers on whether a clinician can appraise and override a discrete output. As systems become faster, less reversible, and more tightly coupled to care processes, oversight must extend beyond output review to include workflow governance: predefined

constraints, escalation rules, traceable control points, and the ability to pause, redirect, or suspend operation before harm propagates⁵. In such settings, mandatory human review becomes especially important when uncertainty is high, data are incomplete, or the consequences of error are substantial^{1,4}.

Without adequate knowledge, time, authority, and technical control, oversight becomes procedural rather than protective. When AI fails, responsibility may collapse onto clinicians expected to catch errors they are not well positioned to detect or correct, a dynamic described as a moral crumple zone⁶. The central question is not whether a human remains in the loop, but whether the organization enables human judgment to prevent harm. Meaningful oversight depends on four conditions: epistemic capacity, cognitive space, decisional authority, and intervention effectiveness.

Human factors research shows why oversight cannot rely on individual vigilance alone. Clinicians can be unduly influenced by AI recommendations, a phenomenon often described as automation bias. In a controlled electronic prescribing study, correct decision support reduced omission errors, whereas incorrect recommendations increased omission errors and were frequently accepted⁷. Similar susceptibility has been observed in mammography, electrocardiogram interpretation, and other clinical settings⁸. Workload, time pressure, interruptions, and competing alerts further reduce vigilance, making it unrealistic to expect clinicians to offset these risks through individual effort alone^{7,9}. Oversight must therefore be designed into the system, not delegated to a single clinician at the point of care.

Beyond cognitive limitations at the point of care, systemic oversight is also necessary because AI may reproduce or amplify inequities embedded in healthcare data, proxy labels, or existing care pathways. Such harms may be difficult to recognize in isolated clinical encounters when model outputs appear consistent with familiar clinical patterns or with assumptions already embedded in practice. Meaningful oversight should therefore include scheduled subgroup performance review, equity-sensitive analysis of concerning output patterns, and escalation pathways for harms that may not be visible to individual clinicians in real time^{10–12}.

Operationalizing meaningful oversight

To move from regulatory expectation to clinical reality, oversight should be treated as a system property that can be designed, assigned, tested, and revised. The four conditions in Table 1 define a minimum operational architecture for asking whether oversight is functionally possible: can a human recognize, question, and alter an AI-mediated trajectory before avoidable harm occurs?

The framework translates the abstract requirement for “human oversight” into actionable questions for governance teams. What must users know about intended use, limitations, and subgroup performance? Does the workflow leave time for independent appraisal? Are clinicians supported to disagree with AI outputs? Can AI-mediated actions be paused, modified, or withdrawn when concerns arise? Framed this way, oversight is not a

Table 1 | Minimum Operational Architecture for Meaningful Oversight of Medical AI

Condition	Operational question	Failure if absent	Required institutional actions	Primary locus of responsibility
Epistemic capacity	Can users judge whether an AI output is reliable, within intended use, and valid for relevant patient subgroups?	Oversight becomes blind trust or blanket rejection, allowing systematic error, hidden bias, or inequitable performance to persist.	Make intended use, limitations, uncertainty, failure modes, and subgroup performance visible; provide model-specific training and onboarding; review subgroup performance and discordance over time.	AI developers/vendors; procurement and governance teams; clinical education
Cognitive space	Does the workflow preserve enough time, attention, and context for independent appraisal?	Time pressure, alert burden, automation bias, and competing demands reduce oversight to perfunctory confirmation.	Monitor alert burden, review latency, and default pathways; avoid compressed decision windows when appraisal is required; design interfaces that support active review rather than passive acceptance.	Clinical departments; operational leaders; workflow owners; interface owners
Decisional authority	Are users supported to question, override, decline, or escalate AI outputs without being treated as noncompliant?	Oversight becomes symbolic, with compliance driven by hierarchy, performance pressure, liability concerns, or unclear expectations for dissent.	Protect justified divergence through policy; define when override, escalation, or mandatory review is expected; review overrides and discordance as safety signals.	Institutional leadership; clinical governance; risk management; departmental leadership
Intervention effectiveness	Can AI-mediated recommendations or actions be paused, corrected, suspended, or deactivated before harm propagates?	Nominal override exists without practical control, allowing errors to spread through clinical, administrative, or patient-facing processes.	Build usable pause, correction, escalation, suspension, and deactivation functions; maintain auditable logs; define thresholds for reconfiguration, restriction, suspension, or decommissioning.	AI developers/vendors; health IT; interface owners; quality and safety governance

This table summarizes the four conditions that form a minimum operational architecture for meaningful oversight of medical AI. The primary locus of responsibility should be adapted to the system's autonomy, reversibility, and downstream coupling. Predictive decision-support tools may require output appraisal and override review, whereas generative or agentic systems that draft, route, or initiate downstream actions require stronger traceability, escalation pathways, and pause or suspension functions.

AI artificial intelligence.

professional virtue expected of individual clinicians. It is a set of clinical, organizational, and technical conditions that can be built and audited.

First, clinicians need epistemic capacity: an understanding of what a model is designed to do, the populations in which it was validated, how performance varies across subgroups, and where failure is most likely. Transparency artifacts such as model cards can help, but they are insufficient without training, onboarding, technical support, and institutional infrastructure¹³. Without that epistemic foundation, clinicians face a false choice between blind trust and blanket rejection of AI, neither of which protects patients or supports sustainable adoption.

Second, clinicians need cognitive space: even well-informed users cannot provide meaningful oversight if the workflow leaves no time to think. Recommendations delivered during compressed decision windows or buried among competing alerts are unlikely to receive careful appraisal. Interface design matters as much as model performance: defaults that make override slower than compliance or present recommendations as mandates push behavior toward automatic acceptance.

Third, clinicians need decisional authority. Clinicians may hesitate to diverge from AI outputs if organizational policy, performance metrics, peer norms, or medicolegal concern implicitly discourage dissent. Oversight becomes symbolic if disagreement is treated as deviance. Institutions should therefore make clear that justified divergence from AI recommendations is expected, documented, and protected¹.

Fourth, intervention must be effective in practice. Some AI outputs, such as automated order sets, triage decisions, or bed assignments, can trigger downstream actions that are difficult to reverse. Claims that a system can always be overridden are hollow if doing so requires extraordinary effort, delays care, or exposes clinicians to scrutiny. High-consequence outputs should therefore be interruptible before irreversible downstream action occurs.

These conditions should be operationalized across the AI lifecycle by assigning responsibility according to control^{1,13}. Developers and vendors should provide documentation on intended use, validation populations, known failure modes, update procedures, interface assumptions, and available override, pause, or stop functions. Procurement and governance teams should assess whether this information is sufficient for local adoption, whether local validation is needed, and whether the system fits institutional risk tolerance. Clinical departments and operational leaders should test the tool in workflow, identify points of time pressure or automation bias, define escalation rules, and ensure model-specific training before use. Health IT and interface owners should make override, correction, routing, and suspension functions usable, auditable, and proportionate to the system's level of actionability. Quality and safety offices should review post-deployment signals, investigate incidents, and be able to recommend reconfiguration, restriction, suspension, or withdrawal. Frontline clinicians should be supported to document overrides, report concerns, and participate in feedback loops, but should not become the default guarantors of safety for systems they cannot meaningfully control.

Concrete oversight safeguards

These responsibilities should be translated into governance, workflow, and design safeguards. Override events should be logged and reviewed as safety signals, not as simple noncompliance. A core oversight dashboard could include override frequency, review latency, AI–human concordance and discordance, subgroup-specific performance drift, patterned discordance across patient groups, and AI-associated adverse events or near misses. These indicators should trigger interpretation rather than serve as standalone scorecards. A high override rate may reflect poor calibration, workflow mismatch, low user trust, or appropriate resistance in difficult cases. A low

override rate may reflect good performance, but also automation bias, time pressure, or weak authority to disagree.

Before deployment, institutions should define who reviews these signals, how often, and which patterns trigger audit or intervention¹³. Review should occur through a multidisciplinary AI oversight or clinical safety process with clinical, technical, workflow, safety, and equity expertise. Follow-up may include case review, subgroup analysis, workflow redesign, interface revision, threshold adjustment, targeted training, temporary restriction, suspension, or decommissioning. Each action should be documented and reassessed to determine whether further control is needed.

Two common deployment scenarios illustrate how these conditions differ across system types: predictive alerting and generative or semi-agentic patient communication^{14,15}.

First, consider a predictive AI tool that generates an inpatient deterioration or sepsis alert. Nominal oversight exists if the alert appears in the electronic health record and a clinician can technically ignore or acknowledge it. Meaningful oversight requires more: users must understand the model's intended use, limitations, uncertainty, and local performance; the alert must arrive when it can be appraised rather than reflexively accepted; clinicians must be supported to disagree when the output conflicts with the clinical picture; and discordant cases or repeated overrides must feed into safety review rather than be treated as individual noncompliance.

Second, consider a generative or semi-agentic system used in patient communication. Such a system might summarize portal messages, draft patient-facing responses, prioritize messages for review, or trigger follow-up tasks such as outreach or scheduling. Here, the oversight problem differs: the clinician may not be reviewing a single recommendation before action is taken; the system may shape what is seen, when it is seen, and which tasks are initiated. Meaningful oversight therefore requires traceable generation and routing, clear responsibility for configuration rules, protected clinician authority to edit, reject, or escalate generated content, and practical controls to pause, redirect, or suspend downstream actions when safety concerns arise.

Oversight also depends on the ability to intervene before harm propagates. Systems that influence clinical action should allow recommendations or downstream actions to be paused, modified, or suspended when concerns arise, with each intervention traceable over time¹³. Without reversible architecture, auditable control points, and feedback loops, oversight remains theoretical. Accountability should therefore match control: clinicians should not carry responsibility for systems they lack the authority, time, or practical means to change.

Conclusion

Human oversight of medical AI is meaningful only when the clinical, organizational, and technical environment enables users to understand, question, and interrupt AI-mediated care. As systems move from recommendations to generated content or initiated care processes, oversight must extend from output review to workflow governance. The framework offered here is a practical governance aid for assigning responsibilities, monitoring safeguards, and revising controls as medical AI evolves.

Data availability

No datasets were generated or analysed during the current study.

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Competing interests

The authors declare no competing interests.

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